

SAIRAM SRIDHARAN

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Work Authorization: F-1 OPT; STEM OPT eligible (3 years total work authorization). No immediate sponsorship required.

EDUCATION

Northeastern University | Boston, MA

May 2026

Masters of Robotics, Electrical and Computer Engineering, **GPA: 3.919**

Related Coursework: Robot Sensing and Navigation, Mobile Robotics, Robot Mechanics and Control, Field Robotics, Assistive Robotics, Reinforcement Learning, Optimization Algorithms, Non-Linear Controls.

Vellore Institute of Technology | Chennai

May 2024

Bachelor of Technology, Mechatronics and Automation, **GPA: 9.12/10**

Related Coursework: Industrial Automation, PLC, Machine Learning, Computer Vision, IoT, Product Design and Development.

TECHNICAL SKILLS

Programming: Python, C++, Rust, MATLAB, SIMULINK

Software and Tools: ROS1/ROS2, Linux, Gazebo, MuJoCo, OpenCV, PyTorch, Docker, Git, SolidWorks, Raspberry Pi, GTSAM

Robotics: SLAM, Visual Odometry, Sensor Fusion, Path Planning, Motion Planning, Computer Vision, Reinforcement Learning

PROJECTS

Autonomous Jenga-Playing Robot | Northeastern University | [Robotic Manipulation – Jenga](#)

- Built a robotic system using the WidowX-250 arm to autonomously play Jenga by integrating perception, control, and mechanical design for real-world manipulation tasks.
- Implemented AprilTag-based block detection, real-time pose transformation, and closed-loop feedback control in ROS 2, achieving successful block extraction without tower collapse.

Camera–LiDAR Extrinsic Calibration On-the-fly | Northeastern University | [Online Camera–LiDAR Calibration](#)

- Developed an online camera–LiDAR extrinsic calibration pipeline using edge-based alignment of projected 3D LiDAR features with 2D image edges.
- Validated robustness through synthetic offset injection and recovery experiments, demonstrating suitability for real-world robotic deployment.

Structure-from-Motion (SfM) 3D Reconstruction Pipeline | Northeastern University | [SfM 3D Reconstruction](#)

- Built a complete SfM pipeline from scratch, including feature extraction, epipolar geometry, pose recovery, triangulation, PnP and global bundle adjustment using GTSAM.
- Validated reconstruction accuracy by comparing outputs with COLMAP and extracting real camera intrinsics (K) and radial distortion (k_1) from COLMAP calibration.

Adaptive Path-Guided RRT (APG-RRT) Implementation | Northeastern University | [APG-RRT Path Planning](#)

- Implemented and evaluated APG-RRT as an efficient alternative to RRT, generating smoother, shorter, and collision-free trajectories in structured environments (corridors, bends, narrow passages).
- Extended the algorithm from 2D Python simulations into a ROS 2 framework, integrating with TurtleBot3 in Gazebo to validate autonomous navigation in realistic robotic scenarios.

Three-Link Planar Biped Walking using Hybrid Zero Dynamics | Northeastern University | [3-Link Biped Control](#)

- Modeled a three-link planar biped and derived hybrid walking dynamics for underactuated locomotion.
- Designed stable walking gaits using virtual constraints, Bézier polynomials, zero dynamics, and feedback linearization.

Route Master: Navigating Dynamic Environments with Advanced Mapping | VIT University | [Autonomous Mobile Robot](#)

- Developed a cost-effective autonomous mobile robot using LiDAR and Raspberry Pi, integrated with ROS for both tele-operated and autonomous navigation.

Performed iterative testing to achieve reliable performance in dynamic indoor environments, with the robot effectively avoiding obstacles and following predefined paths with high precision.

WORK EXPERIENCE

Research Intern | WIKA | Chennai

June 2022 - July 2022

- Assisted the team in design and production of Industrial sensors, including flow, temperature, and pressure measurement devices.
- Contributed to calibrating Industrial sensors to maintain the accuracy and precision of pressure and temperature measurement instruments.

Research Intern | Lucas TVS | Chennai

May 2022 - June 2022

- Monitored and supervised assembly lines for alternators, starters, motors, and wipers used in automotive applications, gaining practical experience in production workflows and process optimization.
- Worked with the R&D department to research product innovations and evaluate emerging technologies for future development.